AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 20 · STUDENT LESSON

Recommendations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 19: recommendations depend on selected preferences.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Apply eligibility requirements before ranking candidates by preferences. Try the recall before opening notes.

Your learning target

I can apply **eligibility** requirements before **ranking** candidates by **preference**s.

I can explain a **recommendation** using its actual scoring rule and limits.

I can interpret feedback in the context of what was shown and distinguish clicks from satisfaction.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

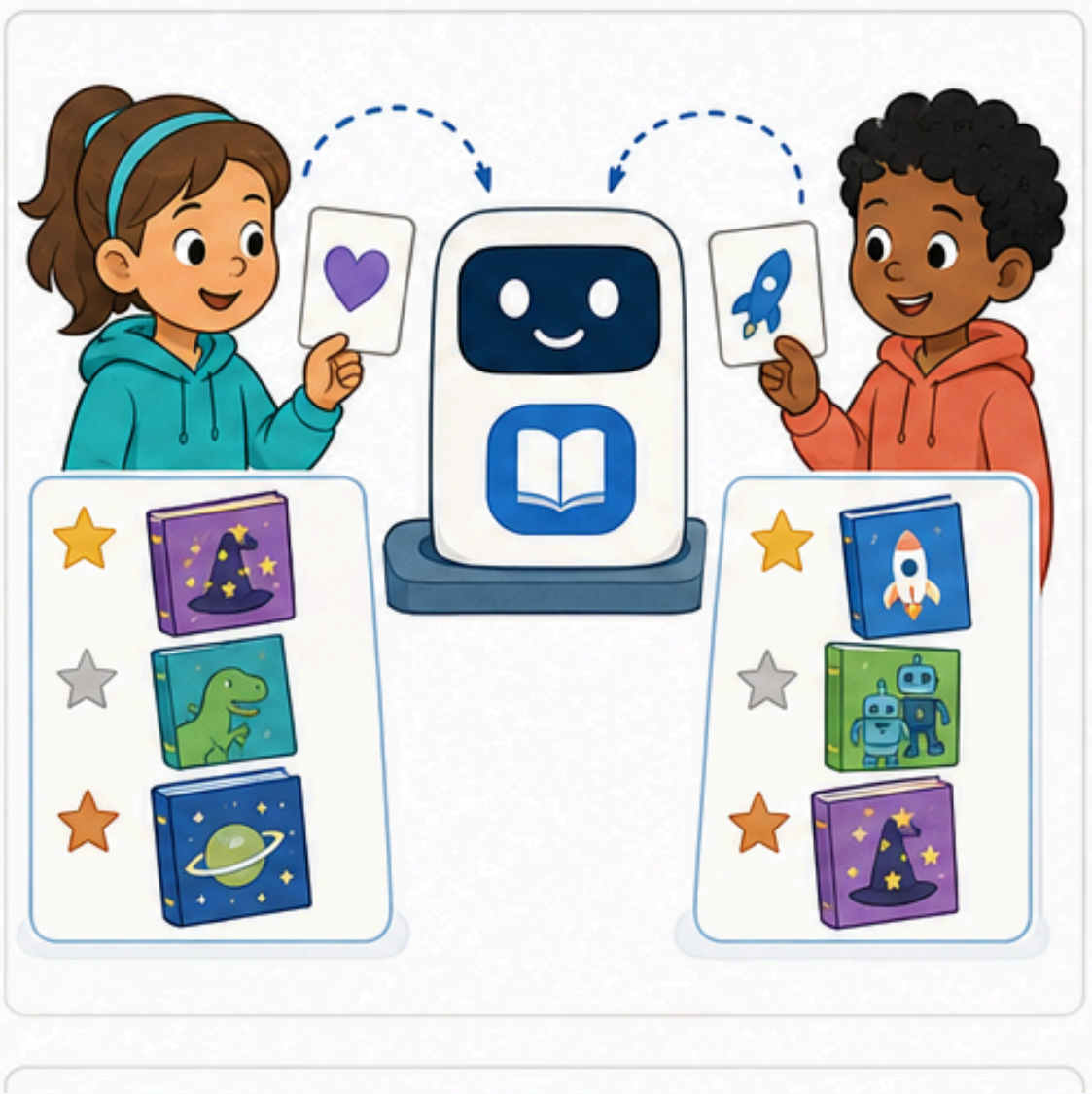
Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A recommender ranks possibilities using goals, information, and **tradeoff**s.



Your target

LEARNING TARGET A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

Learner name:

Readiness for the connected explanation

A book has the highest similarity score but is in a language the reader cannot use. Should the score override the language requirement? Explain.

This week's end goal

By the end, I can filter requirements before **ranking preference**s and avoid overinterpreting clicks.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Recommendations case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY Two readers can receive different book **ranking**s because their goals, reading level, interests, and novelty **preference**s differ.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A **recommendation** suggests an option for a particular purpose. Start with requirements: conditions every suggested item must meet. Then use **preference**s to rank the eligible options.

An item that fails a requirement cannot make up for it by collecting more **preference** points.

A fictional reader needs an available English book. The reader prefers topic space and format comic. BookRec1 gives eligible books 2 points for space and 1 for comic; a known mismatch gives 0.

It returns every eligible book tied for the highest score. If none are eligible, it reports no eligible option rather than ignoring a requirement.

The candidates are J: space, prose, English, available; K: animals, comic, English, available; L: space, comic, Spanish, available; M: space, comic, English, unavailable. These are invented records.

The purpose is to trace a **recommendation**, not to infer anything about real students or collect their personal **preference**s.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Recommendations. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



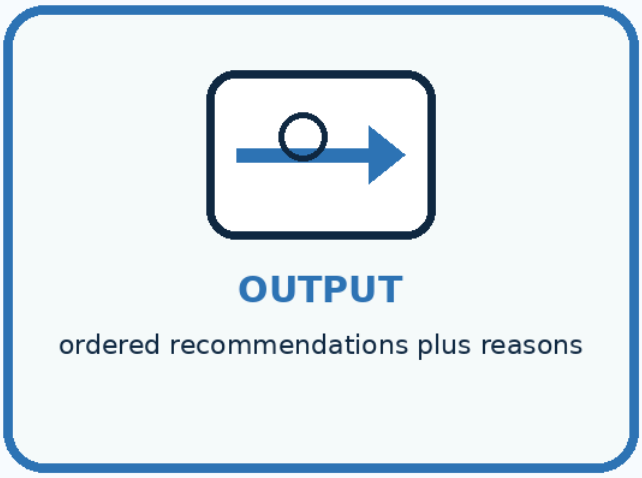
INPUT

reader goals and candidate-book traits



PROCESS

score and rank candidates with chosen tradeoffs



INPUT reader goals and candidate-book traits

PROCESS score and rank candidates with chosen **tradeoff**s

OUTPUT ordered **recommendation**s plus reasons

Key idea

CORE IDEA A recommender ranks possibilities using goals, information, and **tradeoff**s.

How the mechanism works

UNDER THE HOOD **Recommendation** is **ranking**, not merely classification. An **objective** combines relevance signals and constraints; **tradeoff**s may include familiarity, **diversity**, novelty, safety, and availability.

The top score reflects the **objective**, not universal quality.

Draw a fresh example with all three stages:

A closer explanation

Requirements come before preference scores

CANDIDATE	ELIGIBILITY CHECK	RESULT
J: space and prose	English; available	Eligible; 2 points
K: animals and comic	English; available	Eligible; 1 point
L and M	Language or stock fails	Excluded from ranking

BookRec1 recommends J from this candidate set. L and M match both preferences but fail a requirement. The recommendation does not guarantee that the reader will enjoy J.

J passes both requirements and scores 2. K passes and scores 1. L fails the English requirement; M fails availability. Do not place L or M above J because they would earn 3 **preference** points.

They are excluded before the eligible scores are compared.

A faithful explanation says J is available in English and has the highest eligible score because its space topic matches the stated **preference**. It should also mention that J is prose, not comic.

Claiming that the system knows the reader will love J would go beyond the computation and the **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

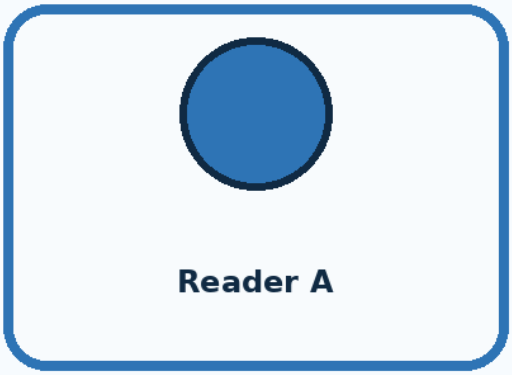
Compare the cases

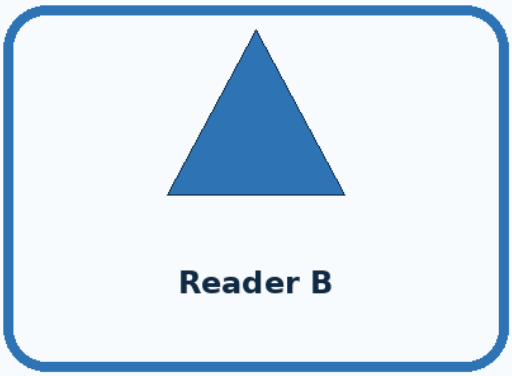
> Compare the supplied cases for Recommendations. Identify the detail that changes the conclusion.

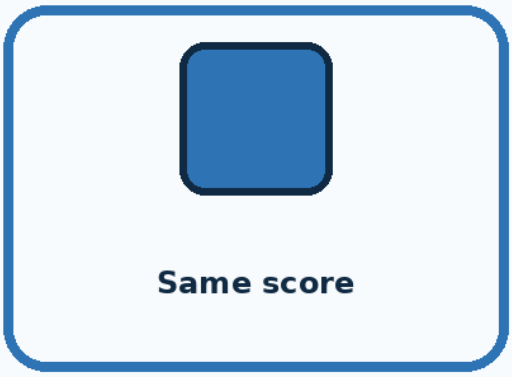
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Reader A Clue: likes survival and long books. Meaning: rank adventure higher.

The case to investigate

CASE 2 — Reader B Clue: wants short humor. Meaning: rank comedy higher.

The case to investigate

CASE 3 — Same score Clue: two books tie. Meaning: need tie policy.

The case to investigate

CASE 4 — Unavailable book Clue: high relevance but unusable. Meaning: constraint changes rank.

Keep this boundary clear

BOUNDARY A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Change the maximum time to 10 minutes.

Predicted result

Prediction: Only C remains eligible and is recommended.

Why the result follows

Reason: B takes **12 minutes**; A is still noisy.

Words that unlock the idea

> Use the vocabulary of Recommendations to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
ranking	ordered list from higher to lower priority	
objective	goal the scoring process tries to satisfy	
tradeoff	gain in one goal that may reduce another	
diversity	meaningful variety among recommended options	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR The highest-ranked item is best for everyone.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Eligibility	Whether an item meets the requirements for consideration
Preference	A desired feature used to rank eligible choices
Recommendation	A suggested option produced under a stated process
Exposure	An opportunity for a person to see an item
Feedback loop	A process in which earlier recommendations influence the feedback used for later recommendations

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STUDENT LESSON · 6 OF 18

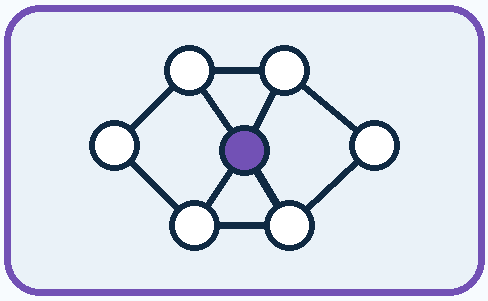
Follow a worked example

> Follow the worked trace for Recommendations, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE Two readers can receive different book **ranking**s because their goals, reading level, interests, and novelty **preference**s differ.

Step 1

STEP 1 — OBSERVE reader goals and candidate-book traits

Step 2

STEP 2 — TRACE score and rank candidates with chosen **tradeoff**s

Step 3

STEP 3 — CONCLUDE ordered **recommendation**s plus reasons

Model answer

MODEL REASONING Use the same transparent scoring method with each reader's stated **preference**s. Different **input**s create different totals. Report **tradeoff**s, ties, and one option outside the top rank to preserve choice.

A claim to avoid

DO NOT OVERCLAIM A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

What new **evidence** could change this answer?

A closer explanation

Recorded event	What it establishes	What it does not establish
J was shown	There was a recorded exposure	The reader noticed or liked it
J was clicked	A recorded interaction occurred	The reader enjoyed the book
Reader said J was too long	A stated reason in this context	All space books are unwanted
K was never shown	No exposure recorded for K	The reader rejected K

Worked explanation · Recommendations

A click can reflect interest, curiosity or a mistaken tap. No click can reflect a missed display, lack of time or a decision not to choose it. Preserve the observed event separately from any interpretation.

More direct feedback may clarify a reason, but one response still describes one person and occasion.

Useful feedback records name the item, what was shown, the system version, the event and any stated reason. Use only information needed for the task.

Our paper exercises use invented feedback; they do not require accounts or recording classmates' browsing behaviour.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Recommendations

A fictional classroom break chooser recommends a suitable paper activity.

The facts we will use

Supplied facts

Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

Supplied facts

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

AI ACADEMY / AI-1/20

Requirements before preferences

	Quiet	Minutes	Score	Status
A	No	8	3	excluded
B	Yes	12	2	eligible
C	Yes	10	1	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/20

Recommendations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Requirements

Reject A because it is noisy despite a preference score of 3.



2. Eligible candidates

B and C both satisfy time and quiet requirements.



3. Preference scores

B scores 2; C scores 1.



4. Recommend

Choose B; preserve the reasons and any later feedback.

CHANGE ONE THING

Change the maximum time to 10 minutes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Requirements	Reject A because it is noisy despite a preference score of 3.
2. Eligible candidates	B and C both satisfy time and quiet requirements.
3. Preference scores	B scores 2; C scores 1.
4. Recommend	Choose B; preserve the reasons and any later feedback.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

What this example does not establish

Keep the stated limit

One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Keep the stated limit

Read the labelled picture: Why does the **recommendation** change without changing anyone's **preference**s?

Keep the stated limit

Change one thing: Change the maximum time to 10 minutes. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Recommendations claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Reader A	likes survival and long books	rank adventure higher

Case	Evidence	What it shows
Reader B	wants short humor	rank comedy higher
Same score	two books tie	need tie policy
Unavailable book	high relevance but unusable	constraint changes rank

Repair two tempting claims

Challenge claim A

CLAIM A The highest-ranked item is best for everyone.

Repair:

Challenge claim B

CLAIM B Personalization removes the need for user choice.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

If feedback says J is too long, length may be a relevant missing feature. Ask what limit matters before adding a requirement.

Do not automatically remove all space books or silently turn one comment into a permanent **preference**. Keep the original feedback and document any change to the **recommendation** rules.

Check the claim against the facts

Check a revised version on reserved paper cases, including ties, no eligible choices and missing required information.

If a required fact is unknown, mark that candidate pending verification and report the limitation; do not assume it passes. If all candidates are pending or excluded, return no verified eligible option.

A **recommendation** leaves the final choice with the reader.

A system recommends only the books it showed last week because those are the only books with clicks. Explain how its own display choices could keep other books out of consideration and name one improvement to the evidence collection

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STUDENT LESSON · 8 OF 18

Practise together

Practise Recommendations with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A recommender ranks possibilities using goals, information, and **tradeoff**s.

Use a fresh example to explain **ranking**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Personalization removes the need for user choice.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose a system shows only space books, then collects clicks only on those books. If it treats those clicks as proof that space is the only wanted topic, it may keep showing space.

Its choices helped produce the **evidence** it later uses. The absence of animal-book clicks says little when animal books were never shown.

> **A log says J was shown and clicked; K was never shown. A designer wants to label J enjoyed and K disliked. Write the two defensible records, identify the unsupported labels, and propose a useful way to ask for feedback without assuming the answer**

A better investigation can ask the fictional reader directly or offer a small set of eligible alternatives with clear choice and optional feedback. Record what was offered as well as the response.

Such a comparison still needs care: order, context and missing responses can affect what is observed. Do not call an unseen item rejected.

Practise with less help: Explain a recommendation honestly

Use workbook W2 for the connected example "**Recommendation**s and feedback". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Recommendation**s and feedback. Proposed explanation: "A high score can cancel a failed requirement". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why does the **recommendation** change without changing anyone's **preference**s?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

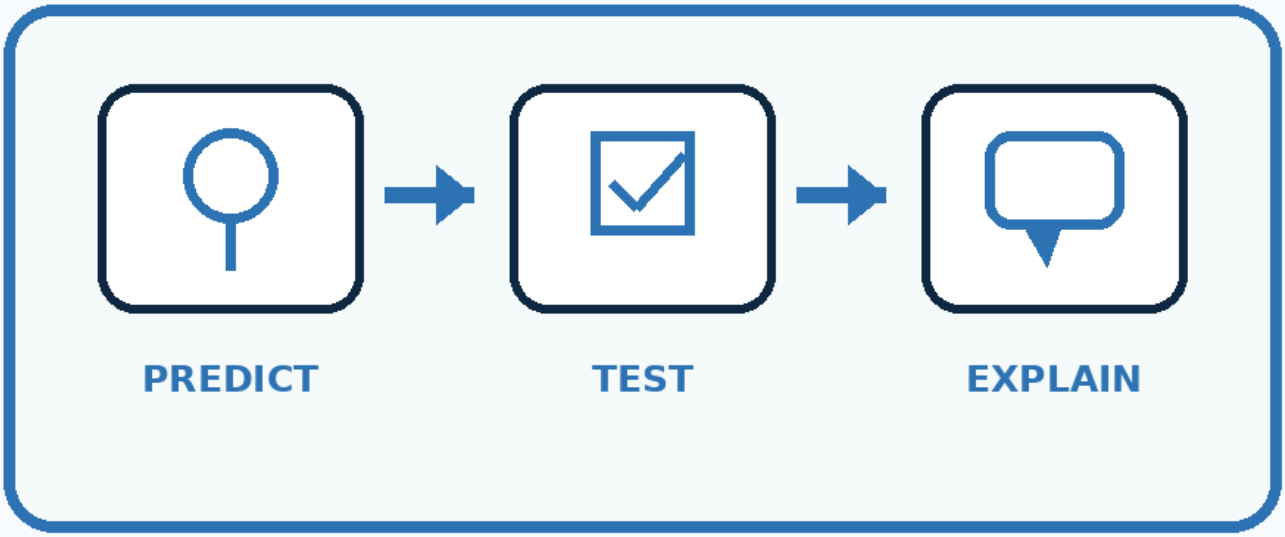
Run the investigation

> Investigate Recommendations. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Two-Reader Book Ranker

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Score books for two readers; explain why **ranking**s differ.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

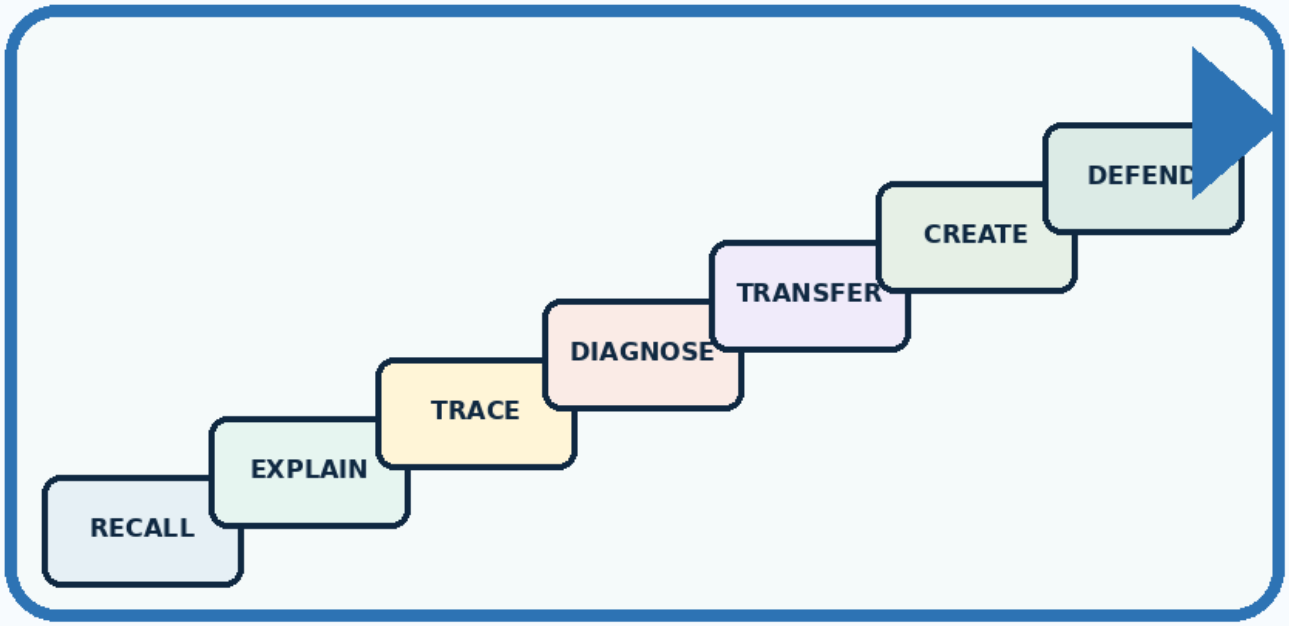
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Recommendations. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **ranking**.

EXPLAIN A recommender ranks possibilities using goals, information, and **tradeoff**s.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: The highest-ranked item is best for everyone.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

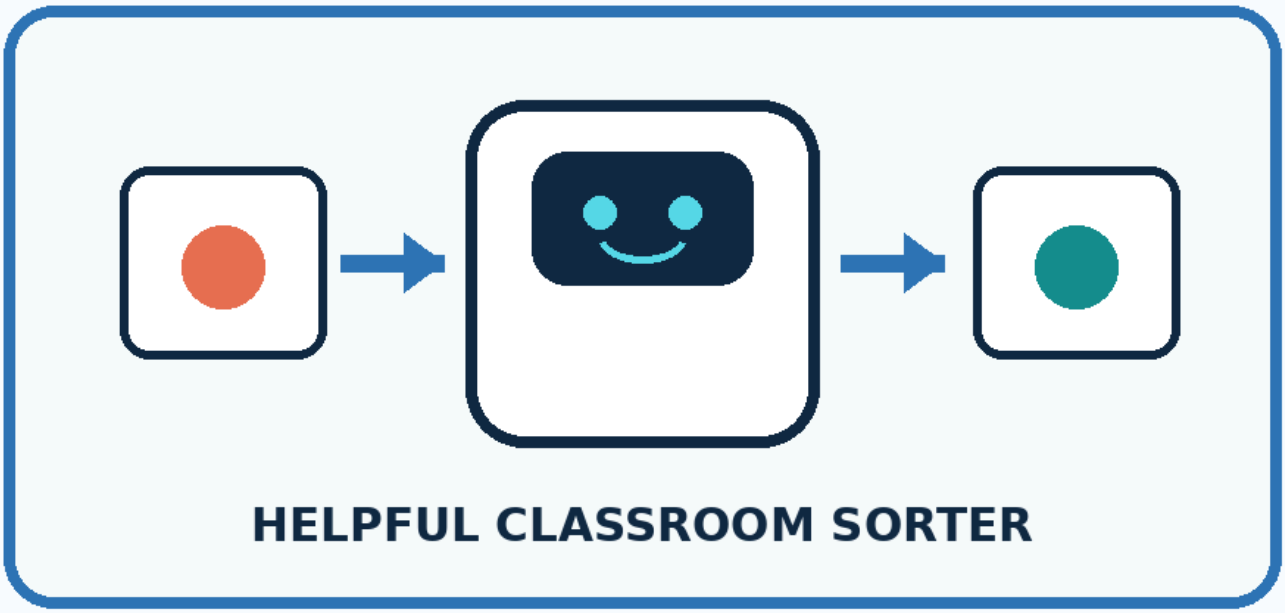
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

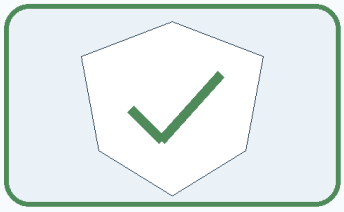
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

My project decision:

Evidence for your project

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

STUDENT LESSON · 12 OF 18

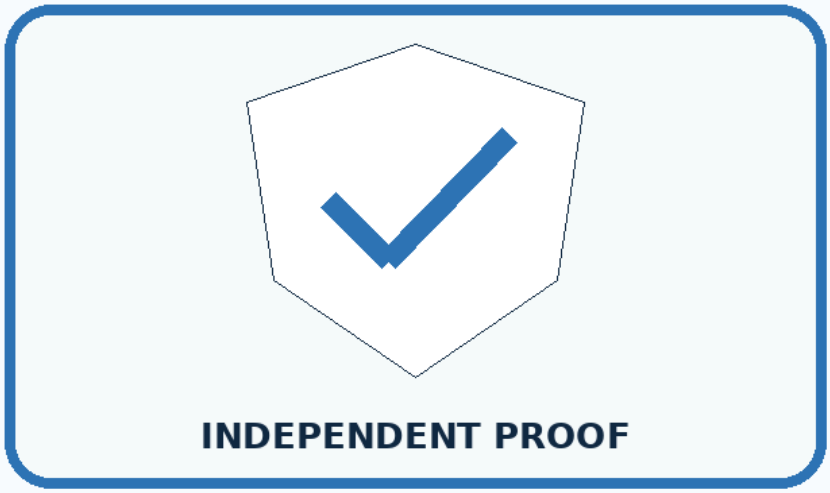
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Score books for two readers, show the arithmetic or rule, and explain why the **ranking**s differ.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Use the same transparent scoring method with each reader's stated **preference**s. Different **input**s create different totals. Report **tradeoff**s, ties, and one option outside the top rank to preserve choice.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Good Data, Bad Data.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Recommendations case for supported practice. Repeated examples do not become fresh assessments.

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A recommender suggests adventure books after one adventure-book choice.

The question

Explain its **evidence** and limitation.

Worked explanation

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Recommendations shape future evidence

Evidence or stage	Value or result
Past choice	One adventure book
Suggestion	More adventure
Feedback	Ask what the reader liked
Next suggestion	Use feedback and offer variety

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer the process to activities

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A recommender suggests a space book because you liked two space books. Does the suggestion prove you will enjoy it? What feedback could help next time?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Recommendations, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional classroom break chooser recommends a suitable paper activity.

By the end, I can	How I will show it
Apply eligibility requirements before ranking candidates by preferences	Explain the deciding step, show a trace or test, and state one limit.
Explain a recommendation using its actual scoring rule and limits	Explain the deciding step, show a trace or test, and state one limit.
Interpret feedback in the context of what was shown and distinguish clicks from satisfaction	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Recommendations without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Apply **eligibility** requirements before **ranking** candidates by **preference**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom break chooser recommends a suitable paper activity.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Choose B; preserve the reasons and any later feedback.

1. Recommend

2. Requirements

3. Eligible candidates

4. **Preference** scores

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Recommend Choose B; preserve the reasons and any later feedback. A hard requirement determines whether an option can be chosen at all.

Preferences rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the maximum time to 10 minutes. Which conclusion is supported by the supplied facts?

1. Only C remains eligible and is recommended.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Only C remains eligible and is recommended. B takes **12 minutes**; A is still noisy.

Repair a missing building block

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

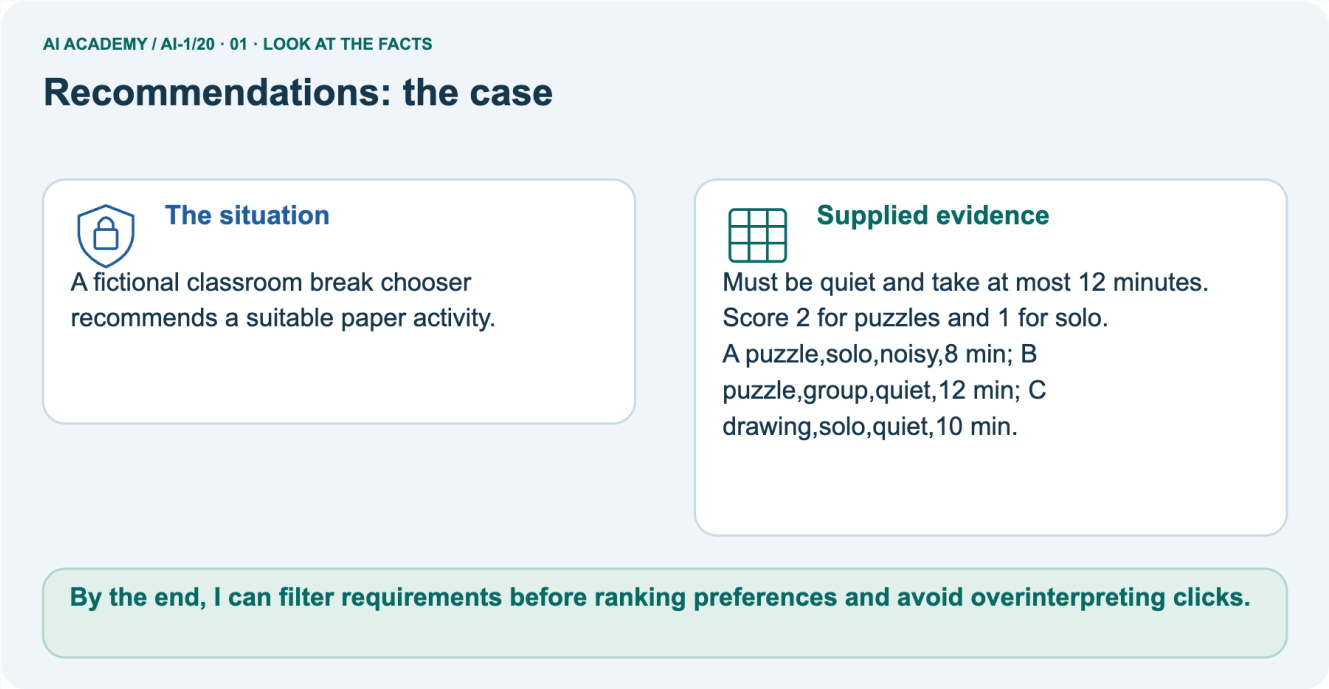
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Recommendations graphic using its labels, deciding evidence and relevant limitation.

01. Recommendations: the case



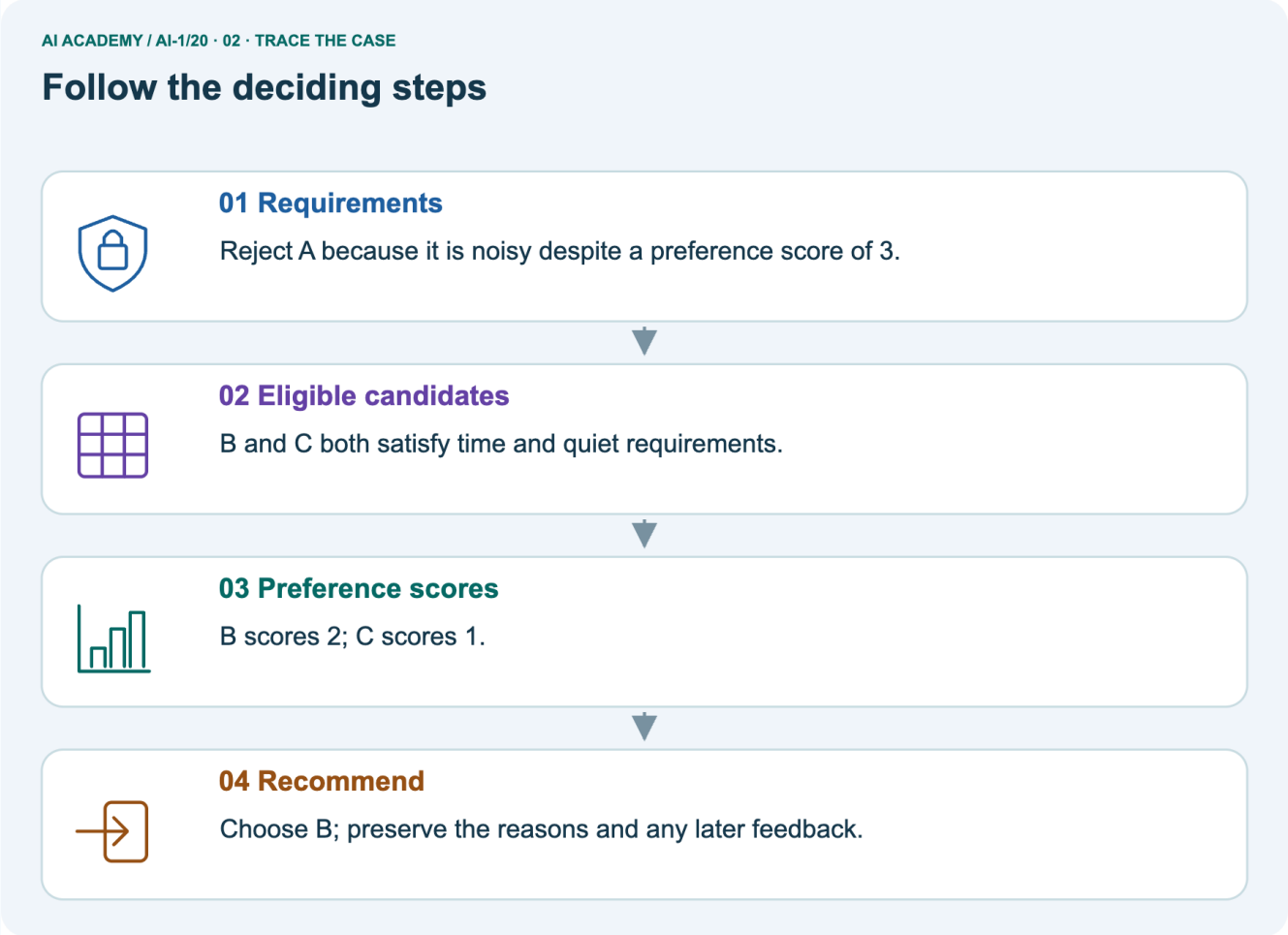
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom break chooser recommends a suitable paper activity. Supplied facts: Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

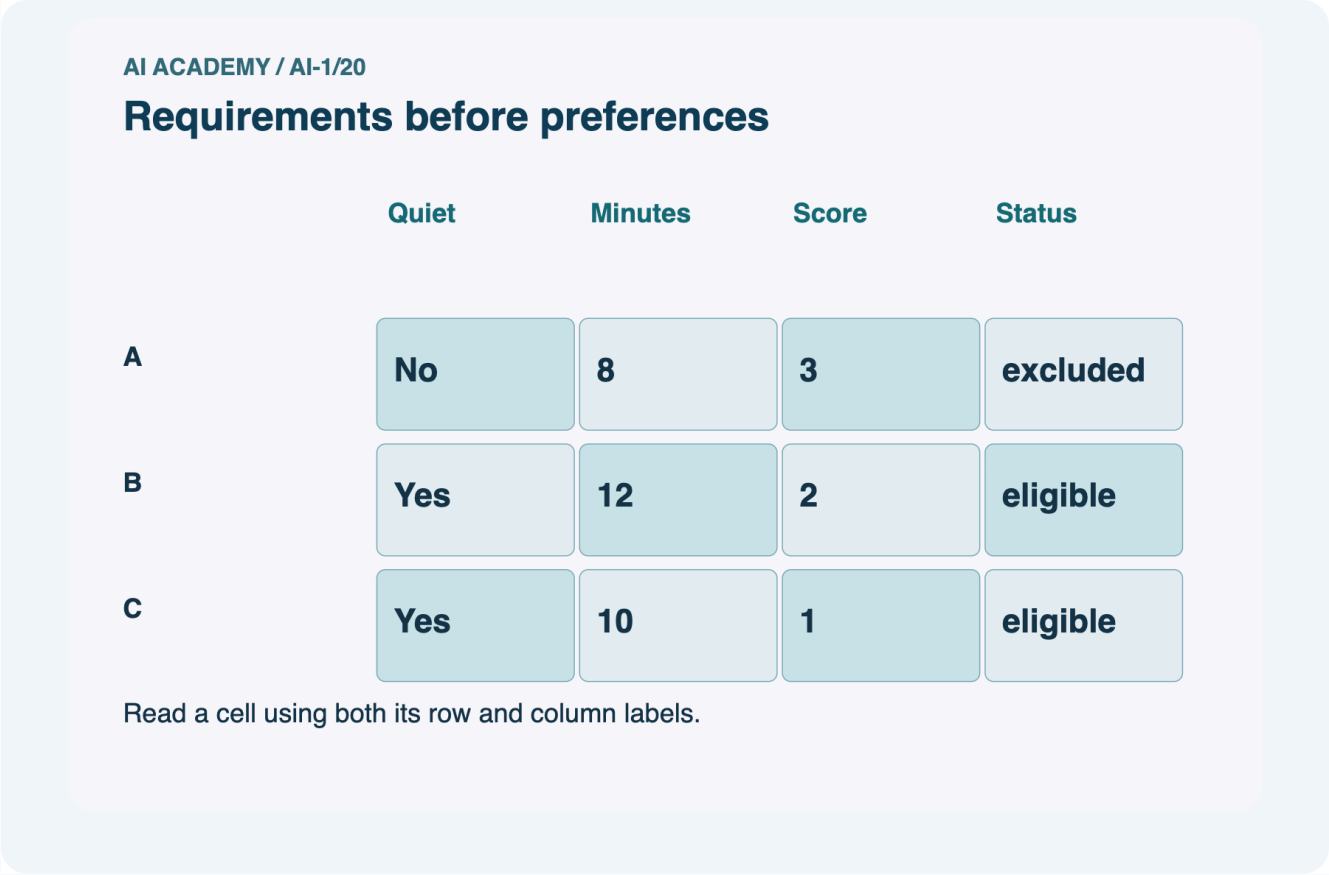


Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements: Reject A because it is noisy despite a **preference** score of 3. → Eligible candidates: B and C both satisfy time and quiet requirements. → **Preference** scores: B scores 2; C scores 1. → Recommend: Choose B; preserve the reasons and any later feedback.

Point to each numbered stage and explain which recorded **input** it uses.

03. Requirements before preferences

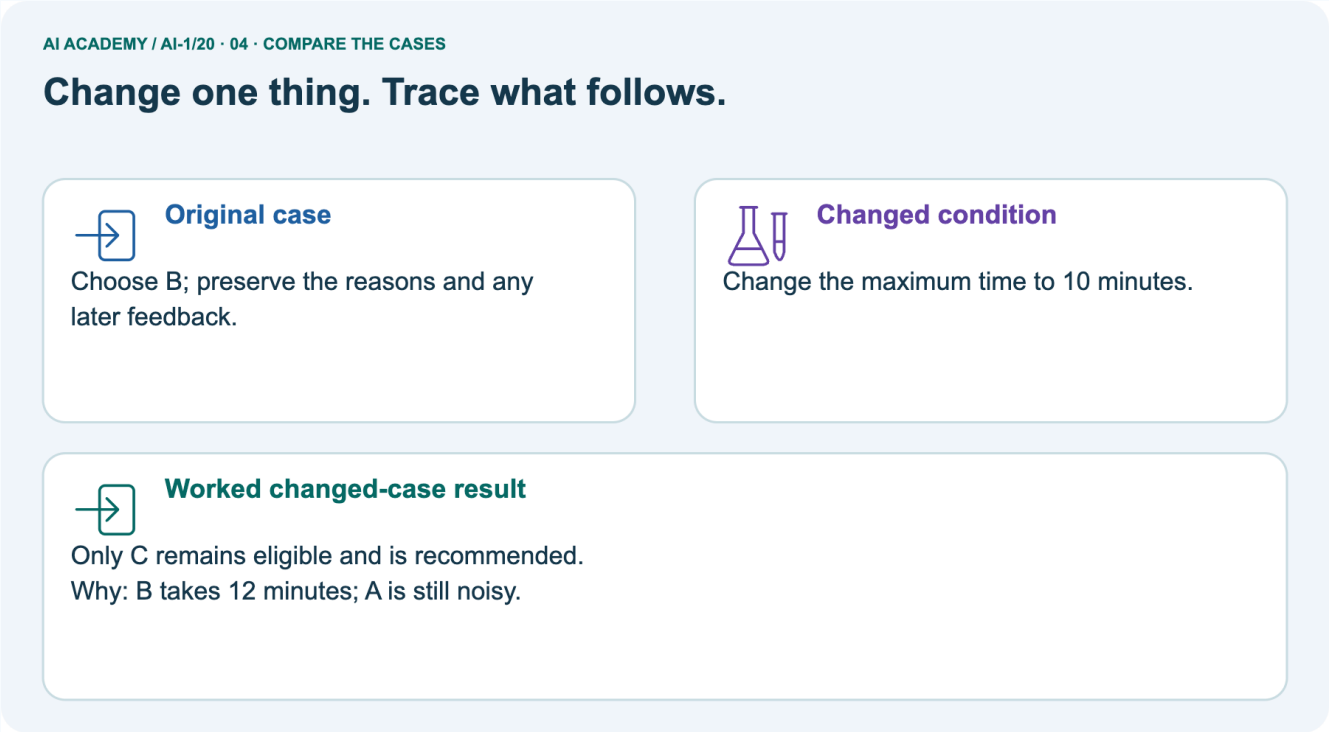


Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements before **preference**s. A: Quiet = No, Minutes = 8, Score = 3, Status = excluded. B: Quiet = Yes, Minutes = 12, Score = 2, Status = eligible. C: Quiet = Yes, Minutes = 10, Score = 1, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



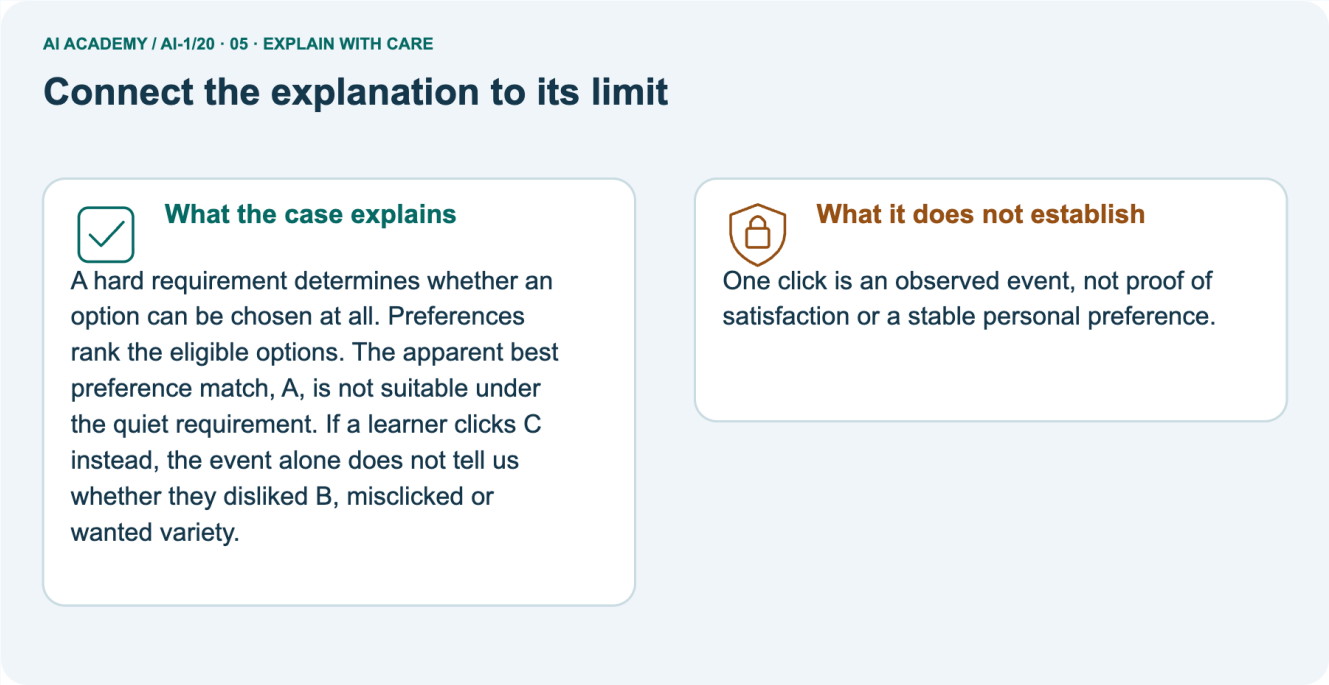
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Choose B; preserve the reasons and any later feedback. Changed condition: Change the maximum time to 10 minutes. Expected result: Only C remains eligible and is recommended.

Reason: B takes **12 minutes**; A is still noisy.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



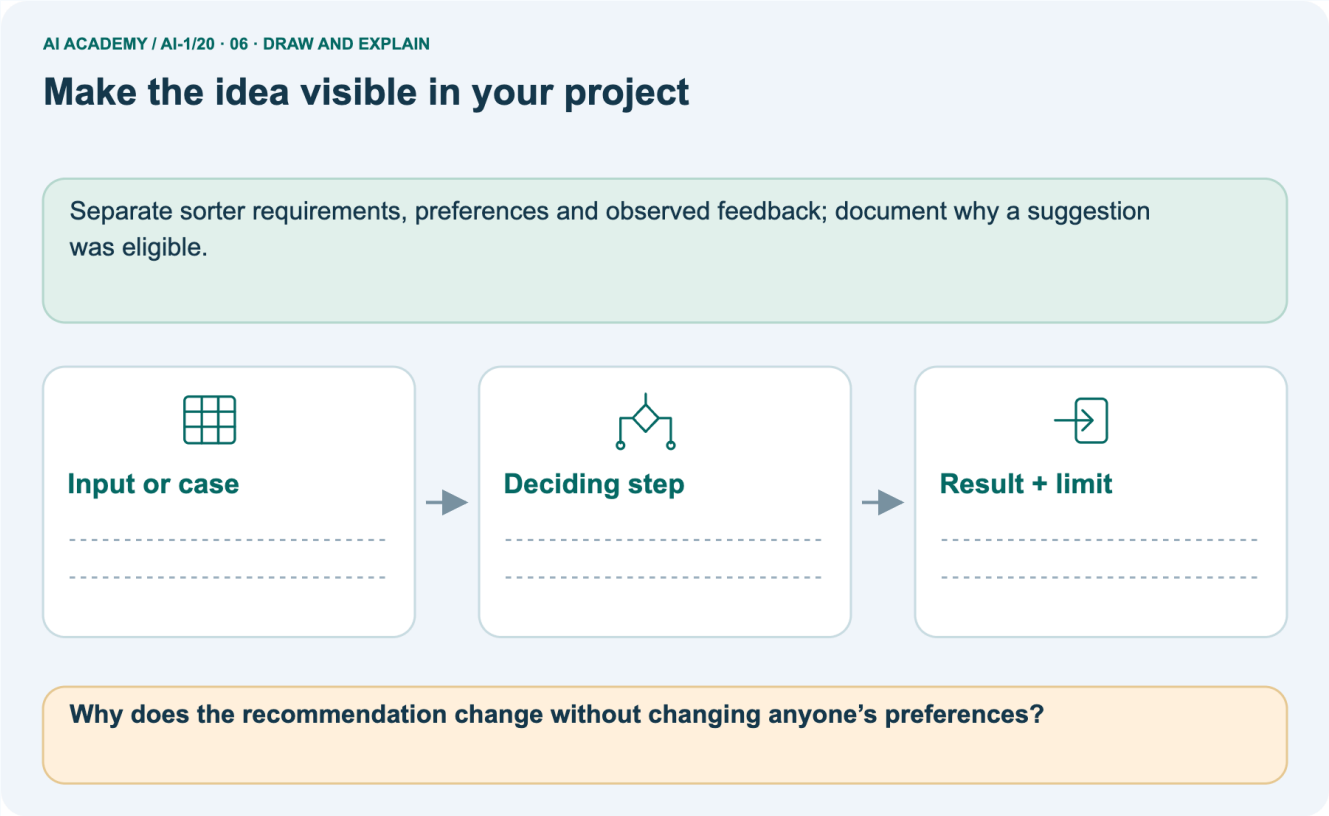
Follow the labelled picture with the matching explanation. Open the vector diagram

A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety. Limit: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible. Draw your **input**, deciding step, result and limit.

Supported question: Why does the **recommendation** change without changing anyone’s **preference**s?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.